

Week 11

Linear Algebra

By

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4.5

Coordinates and Basis

We usually think of a line as being one-dimensional, a plane as two-dimensional, and the space around us as three-dimensional. It is the primary goal of this section and the next to make this intuitive notion of dimension precise. In this section we will discuss coordinate systems in general vector spaces and lay the groundwork for a precise definition of dimension in the next section.

Basis for a Vector Space

Our next goal is to extend the concepts of “basis vectors” and “coordinate systems” to general vector spaces, and for that purpose we will need some definitions. Vector spaces fall into two categories: A vector space V is said to be **finite-dimensional** if there is a finite set of vectors in V that spans V and is said to be **infinite-dimensional** if no such set exists.

Definition 1

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a set of vectors in a finite-dimensional vector space V , then S is called a **basis** for V if:

- (a) S spans V .
- (b) S is linearly independent.

EXAMPLE 3 | Another Basis for R^3

Show that the vectors $\mathbf{v}_1 = (1, 2, 1)$, $\mathbf{v}_2 = (2, 9, 0)$, and $\mathbf{v}_3 = (3, 3, 4)$ form a basis for R^3 .

Solution We must show that these vectors are linearly independent and span R^3 . To prove linear independence we must show that the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0} \quad (1)$$

has only the trivial solution; and to prove that the vectors span R^3 we must show that every vector $\mathbf{b} = (b_1, b_2, b_3)$ in R^3 can be expressed as

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{b} \quad (2)$$

By equating corresponding components on the two sides, these two equations can be expressed as the linear systems

$$\begin{array}{rcl} c_1 + 2c_2 + 3c_3 = 0 & & c_1 + 2c_2 + 3c_3 = b_1 \\ 2c_1 + 9c_2 + 3c_3 = 0 & \text{and} & 2c_1 + 9c_2 + 3c_3 = b_2 \\ c_1 & & + 4c_3 = 0 & c_1 & & + 4c_3 = b_3 \end{array} \quad (3)$$

(verify). Thus, we have reduced the problem to showing that in (3) the homogeneous system has only the trivial solution and that the nonhomogeneous system is consistent for all values of b_1, b_2 , and b_3 . But the two systems have the same coefficient matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 9 & 3 \\ 1 & 0 & 4 \end{bmatrix}$$

so it follows from parts (b), (e), and (g) of Theorem 2.3.8 that we can prove both results at the same time by showing that $\det(A) \neq 0$. We leave it for you to confirm that $\det(A) = -1$, which proves that the vectors $\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$ form a basis for R^3 .

EXAMPLE 4 | The Standard Basis for M_{mn}

Show that the matrices

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad M_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad M_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

form a basis for the vector space M_{22} of 2×2 matrices.

Solution We must show that the matrices are linearly independent and span M_{22} . To prove linear independence we must show that the equation

$$c_1M_1 + c_2M_2 + c_3M_3 + c_4M_4 = \mathbf{0} \tag{4}$$

has only the trivial solution, where $\mathbf{0}$ is the 2×2 zero matrix; and to prove that the matrices span M_{22} we must show that every 2×2 matrix

$$B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

can be expressed as

$$c_1 M_1 + c_2 M_2 + c_3 M_3 + c_4 M_4 = B \quad (5)$$

The matrix forms of Equations (4) and (5) are

$$c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

and

$$c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

which can be rewritten as

$$\begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

which can be rewritten as

$$\begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Since the first equation has only the trivial solution

$$c_1 = c_2 = c_3 = c_4 = 0$$

the matrices are linearly independent, and since the second equation has the solution

$$c_1 = a, \quad c_2 = b, \quad c_3 = c, \quad c_4 = d$$

the matrices span M_{22} . This proves that the matrices M_1, M_2, M_3, M_4 form a basis for M_{22} . More generally, the mn different matrices whose entries are zero except for a single entry of 1 form a basis for M_{mn} called the ***standard basis for M_{mn}*** .

Coordinates Relative to a Basis

Earlier in this section we drew an informal analogy between basis vectors and coordinate systems. Our next goal is to make this informal idea precise by defining the notion of a coordinate system in a general vector space. The following theorem will be our first step in that direction.

Definition 2

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an ordered basis for a vector space V , and

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n$$

is the expression for a vector $\mathbf{v}$ in terms of the basis S , then the scalars $c_1, c_2, \dots, c_n$ are called the ***coordinates of $\mathbf{v}$ relative to the basis S*** . The vector $(c_1, c_2, \dots, c_n)$ in R^n constructed from these coordinates is called the ***coordinate vector of $\mathbf{v}$ relative to S*** ; it is denoted by

$$(\mathbf{v})_S = (c_1, c_2, \dots, c_n) \tag{6}$$

EXAMPLE 9 | Coordinates in R^3

(a) We showed in Example 3 that the vectors

$$\mathbf{v}_1 = (1, 2, 1), \quad \mathbf{v}_2 = (2, 9, 0), \quad \mathbf{v}_3 = (3, 3, 4)$$

form a basis for R^3 . Find the coordinate vector of $\mathbf{v} = (5, -1, 9)$ relative to the basis $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

(b) Find the vector $\mathbf{v}$ in R^3 whose coordinate vector relative to S is $(\mathbf{v})_S = (-1, 3, 2)$.

Solution (a) To find $(\mathbf{v})_S$ we must first express $\mathbf{v}$ as a linear combination of the vectors in S ; that is, we must find values of c_1 , c_2 , and c_3 such that

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$$

or, in terms of components,

$$(5, -1, 9) = c_1(1, 2, 1) + c_2(2, 9, 0) + c_3(3, 3, 4)$$

Equating corresponding components gives

$$c_1 + 2c_2 + 3c_3 = 5$$

$$2c_1 + 9c_2 + 3c_3 = -1$$

$$c_1 + 4c_3 = 9$$

Solving this system we obtain $c_1 = 1$, $c_2 = -1$, $c_3 = 2$ (verify). Therefore,

$$(\mathbf{v})_S = (1, -1, 2)$$

Solution (b) Using the definition of $(\mathbf{v})_S$, we obtain

$$\begin{aligned}\mathbf{v} &= (-1)\mathbf{v}_1 + 3\mathbf{v}_2 + 2\mathbf{v}_3 \\ &= (-1)(1, 2, 1) + 3(2, 9, 0) + 2(3, 3, 4) = (11, 31, 7)\end{aligned}$$

Let's solve the system of equations shown in the image step by step:

$$(1) \quad c_1 + 2c_2 + 3c_3 = 5$$

$$(2) \quad 2c_1 + 9c_2 + 3c_3 = -1$$

$$(3) \quad c_1 + 0c_2 + 4c_3 = 9$$

Step 1: Use equation (3) to express c_1

From equation (3):

$$c_1 + 4c_3 = 9 \Rightarrow c_1 = 9 - 4c_3 \quad (4)$$

Step 2: Substitute (4) into (1) and (2)

Substitute into (1):

$$\begin{aligned}(9 - 4c_3) + 2c_2 + 3c_3 &= 5 \\ \Rightarrow 9 + 2c_2 - c_3 &= 5 \\ \Rightarrow 2c_2 - c_3 &= -4 \quad (5)\end{aligned}$$

Substitute into (2):

$$\begin{aligned}2(9 - 4c_3) + 9c_2 + 3c_3 &= -1 \\ \Rightarrow 18 - 8c_3 + 9c_2 + 3c_3 &= -1 \\ \Rightarrow 9c_2 - 5c_3 &= -19 \quad (6)\end{aligned}$$

Step 3: Solve equations (5) and (6)

From (5):

$$2c_2 = c_3 - 4 \Rightarrow c_2 = \frac{c_3 - 4}{2} \quad (7)$$

Substitute (7) into (6):

$$\begin{aligned}
& 9 \left(\frac{c_3 - 4}{2} \right) - 5c_3 = -19 \\
& \Rightarrow \frac{9c_3 - 36}{2} - 5c_3 = -19 \\
& \Rightarrow \frac{9c_3 - 36 - 10c_3}{2} = -19 \\
& \Rightarrow \frac{-c_3 - 36}{2} = -19 \\
& \Rightarrow -c_3 - 36 = -38 \\
& \Rightarrow c_3 = 2
\end{aligned}$$

Step 4: Back-substitute to find c_2 and c_1

From (7):

$$c_2 = \frac{2 - 4}{2} = -1$$

From (4):

From (4):

$$c_1 = 9 - 4(2) = 1$$

$$c_1 = 1, \quad c_2 = -1, \quad c_3 = 2$$

Exercise Set 4.5

1. Use the method of Example 3 to show that the following set of vectors forms a basis for $\mathbb{R}^2$.

$$\{(2, 1), (3, 0)\}$$

2. Use the method of Example 3 to show that the following set of vectors forms a basis for $\mathbb{R}^3$.

$$\{(3, 1, -4), (2, 5, 6), (1, 4, 8)\}$$

3. Show that the following polynomials form a basis for P_2 .

$$x^2 + 1, \quad x^2 - 1, \quad 2x - 1$$

4. Show that the following polynomials form a basis for P_3 .

$$1 + x, \quad 1 - x, \quad 1 - x^2, \quad 1 - x^3$$

5. Show that the following matrices form a basis for M_{22} .

$$\begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix}, \quad \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$$

6. Show that the following matrices form a basis for M_{22} .

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

7. In each part, show that the set of vectors is not a basis for R^3 .

a. $\{(2, -3, 1), (4, 1, 1), (0, -7, 1)\}$

b. $\{(1, 6, 4), (2, 4, -1), (-1, 2, 5)\}$

8. Show that the following vectors do not form a basis for P_2 .

$$1 - 3x + 2x^2, \quad 1 + x + 4x^2, \quad 1 - 7x$$

9. Show that the following matrices do not form a basis for M_{22} .

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad \begin{bmatrix} 2 & -2 \\ 3 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$$

- 11.** Find the coordinate vector of $\mathbf{w}$ relative to the basis $S = \{\mathbf{u}_1, \mathbf{u}_2\}$ for R^2 .
- a.** $\mathbf{u}_1 = (2, -4)$, $\mathbf{u}_2 = (3, 8)$; $\mathbf{w} = (1, 1)$
 - b.** $\mathbf{u}_1 = (1, 1)$, $\mathbf{u}_2 = (0, 2)$; $\mathbf{w} = (a, b)$
- 12.** Find the coordinate vector of $\mathbf{w}$ relative to the basis $S = \{\mathbf{u}_1, \mathbf{u}_2\}$ for R^2 .
- a.** $\mathbf{u}_1 = (1, -1)$, $\mathbf{u}_2 = (1, 1)$; $\mathbf{w} = (1, 0)$
 - b.** $\mathbf{u}_1 = (1, -1)$, $\mathbf{u}_2 = (1, 1)$; $\mathbf{w} = (0, 1)$
- 13.** Find the coordinate vector of $\mathbf{v}$ relative to the basis $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for R^3 .
- a.** $\mathbf{v} = (2, -1, 3)$; $\mathbf{v}_1 = (1, 0, 0)$, $\mathbf{v}_2 = (2, 2, 0)$,
 $\mathbf{v}_3 = (3, 3, 3)$
 - b.** $\mathbf{v} = (5, -12, 3)$; $\mathbf{v}_1 = (1, 2, 3)$, $\mathbf{v}_2 = (-4, 5, 6)$,
 $\mathbf{v}_3 = (7, -8, 9)$

14. Find the coordinate vector of $\mathbf{p}$ relative to the basis $S = \{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ for P_2 .

a. $\mathbf{p} = 4 - 3x + x^2$; $\mathbf{p}_1 = 1$, $\mathbf{p}_2 = x$, $\mathbf{p}_3 = x^2$

b. $\mathbf{p} = 2 - x + x^2$; $\mathbf{p}_1 = 1 + x$, $\mathbf{p}_2 = 1 + x^2$, $\mathbf{p}_3 = x + x^2$

In Exercises 15–16, first show that the set $S = \{A_1, A_2, A_3, A_4\}$ is a basis for M_{22} , then express A as a linear combination of the vectors in S , and then find the coordinate vector of A relative to S .

15. $A_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $A_2 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, $A_3 = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$,

$$A_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}; \quad A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\mathbf{16.} \quad A_1 = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$A_4 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}; \quad A = \begin{bmatrix} 6 & 2 \\ 5 & 3 \end{bmatrix}$$

In Exercises 17–18, first show that the set $S = \{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ is a basis for P_2 , then express $\mathbf{p}$ as a linear combination of the vectors in S , and then find the coordinate vector of $\mathbf{p}$ relative to S .

$$\mathbf{17.} \quad \mathbf{p}_1 = 1 + x + x^2, \quad \mathbf{p}_2 = x + x^2, \quad \mathbf{p}_3 = x^2;$$

$$\mathbf{p} = 7 - x + 2x^2$$

$$\mathbf{18.} \quad \mathbf{p}_1 = 1 + 2x + x^2, \quad \mathbf{p}_2 = 2 + 9x, \quad \mathbf{p}_3 = 3 + 3x + 4x^2;$$

$$\mathbf{p} = 2 + 17x - 3x^2$$

1. Vectors $(2,1)$ and $(3,0)$ are linearly independent if the vector equation

$$c_1(2,1) + c_2(3,0) = (0,0)$$

has only the trivial solution. For these vectors to span R^2 , it must be possible to express every vector $\mathbf{b} = (b_1, b_2)$ in R^2 as

$$c_1(2,1) + c_2(3,0) = (b_1, b_2)$$

These two equations can be rewritten as linear systems

$$\begin{array}{rclcl} 2c_1 & + & 3c_2 & = & 0 \\ c_1 & & & = & 0 \end{array} \quad \text{and} \quad \begin{array}{rclcl} 2c_1 & + & 3c_2 & = & b_1 \\ c_1 & & & = & b_2 \end{array}$$

Since the coefficient matrix of both systems has determinant $\begin{vmatrix} 2 & 3 \\ 1 & 0 \end{vmatrix} = -3 \neq 0$, it follows from parts (b), (e), and (g) of Theorem 2.3.8 that the homogeneous system has only the trivial solution and the nonhomogeneous system is consistent for all real values b_1 and b_2 . Therefore the vectors $(2,1)$ and $(3,0)$ are linearly independent and span R^2 so that they form a basis for R^2 .

2. Vectors $(3, 1, -4)$, $(2, 5, 6)$, and $(1, 4, 8)$ are linearly independent if the vector equation

$$c_1(3, 1, -4) + c_2(2, 5, 6) + c_3(1, 4, 8) = (0, 0, 0)$$

has only the trivial solution. For these vectors to span R^3 , it must be possible to express every vector $\mathbf{b} = (b_1, b_2, b_3)$ in R^3 as

$$c_1(3, 1, -4) + c_2(2, 5, 6) + c_3(1, 4, 8) = (b_1, b_2, b_3)$$

These two equations can be rewritten as linear systems

$$\begin{array}{rrrrrr} 3c_1 & + & 2c_2 & + & 1c_3 & = & 0 \\ 1c_1 & + & 5c_2 & + & 4c_3 & = & 0 \\ -4c_1 & + & 6c_2 & + & 8c_3 & = & 0 \end{array} \quad \text{and} \quad \begin{array}{rrrrrr} 3c_1 & + & 2c_2 & + & 1c_3 & = & b_1 \\ 1c_1 & + & 5c_2 & + & 4c_3 & = & b_2 \\ -4c_1 & + & 6c_2 & + & 8c_3 & = & b_3 \end{array}$$

Since the coefficient matrix of both systems has determinant $\begin{vmatrix} 3 & 2 & 1 \\ 1 & 5 & 4 \\ -4 & 6 & 8 \end{vmatrix} = 26 \neq 0$, it follows from

parts (b), (e), and (g) of Theorem 2.3.8 that the homogeneous system has only the trivial solution and the nonhomogeneous system is consistent for all real values b_1 , b_2 , and b_3 . Therefore the vectors $(3, 1, -4)$, $(2, 5, 6)$, and $(1, 4, 8)$ are linearly independent and span R^3 so that they form a basis for R^3 .

3. Polynomials $x^2 + 1$, $x^2 - 1$, and $2x - 1$ are linearly independent if the equation

$$c_1(x^2 + 1) + c_2(x^2 - 1) + c_3(2x - 1) = 0$$

has only the trivial solution. For these polynomials to span P_2 , it must be possible to express every polynomial $a_0 + a_1x + a_2x^2$ as

$$c_1(x^2 + 1) + c_2(x^2 - 1) + c_3(2x - 1) = a_0 + a_1x + a_2x^2$$

Grouping the terms on the left hand side of both equations as $(c_1 - c_2 - c_3) + (2c_3)x + (c_1 + c_2)x^2$ these equations can be rewritten as linear systems

$$\begin{array}{rclcl} 1c_1 & - & 1c_2 & - & 1c_3 & = & 0 \\ 0c_1 & + & 0c_2 & + & 2c_3 & = & 0 \\ 1c_1 & + & 1c_2 & + & 0c_3 & = & 0 \end{array} \quad \text{and} \quad \begin{array}{rclcl} 1c_1 & - & 1c_2 & - & 1c_3 & = & a_0 \\ 0c_1 & + & 0c_2 & + & 2c_3 & = & a_1 \\ 1c_1 & + & 1c_2 & + & 0c_3 & = & a_2 \end{array}$$

Since the coefficient matrix of both systems has determinant $\begin{vmatrix} 1 & -1 & -1 \\ 0 & 0 & 2 \\ 1 & 1 & 0 \end{vmatrix} = -4 \neq 0$, it follows from

parts (b), (e), and (g) of Theorem 2.3.8 that the homogeneous system has only the trivial solution and the nonhomogeneous system is consistent for all real values a_0 , a_1 , and a_2 . Therefore the polynomials $x^2 + 1$, $x^2 - 1$, and $2x - 1$ are linearly independent and span P_2 so that they form a basis for P_2 .

4. Polynomials $1 + x$, $1 - x$, $1 - x^2$, and $1 - x^3$ are linearly independent if the equation

$$c_1(1 + x) + c_2(1 - x) + c_3(1 - x^2) + c_4(1 - x^3) = 0$$

has only the trivial solution. For these polynomials to span P_3 , it must be possible to express every polynomial $a_0 + a_1x + a_2x^2 + a_3x^3$ as

$$c_1(1 + x) + c_2(1 - x) + c_3(1 - x^2) + c_4(1 - x^3) = a_0 + a_1x + a_2x^2 + a_3x^3$$

Grouping the terms on the left hand side of both equations as

$(c_1 + c_2 + c_3 + c_4) + (c_1 - c_2)x - c_3x^2 - c_4x^3$ these equations can be rewritten as linear systems

$$\begin{array}{rrrrrr} 1c_1 & + & 1c_2 & + & 1c_3 & + & 1c_4 & = & 0 \\ 1c_1 & - & 1c_2 & + & 0c_3 & + & 0c_4 & = & 0 \\ 0c_1 & + & 0c_2 & - & 1c_3 & + & 0c_4 & = & 0 \\ 0c_1 & + & 0c_2 & + & 0c_3 & - & 1c_4 & = & 0 \end{array} \quad \text{and} \quad \begin{array}{rrrrrr} 1c_1 & + & 1c_2 & + & 1c_3 & + & 1c_4 & = & a_0 \\ 1c_1 & - & 1c_2 & + & 0c_3 & + & 0c_4 & = & a_1 \\ 0c_1 & + & 0c_2 & - & 1c_3 & + & 0c_4 & = & a_2 \\ 0c_1 & + & 0c_2 & + & 0c_3 & - & 1c_4 & = & a_3 \end{array}$$

Since the coefficient matrix of both systems has determinant $\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{vmatrix} = -2 \neq 0$, it follows

from parts (b), (e), and (g) of Theorem 2.3.8 that the homogeneous system has only the trivial solution and the nonhomogeneous system is consistent for all real values a_0, a_1, a_2 , and a_3 .

Therefore the polynomials $1 + x$, $1 - x$, $1 - x^2$ and $1 - x^3$ are linearly independent and span P_3 so that they form a basis for P_3 .

5. Matrices $\begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix}$, $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix}$, and $\begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$ are linearly independent if the equation

$$c_1 \begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix} + c_2 \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix} + c_4 \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

has only the trivial solution. For these matrices to span M_{22} , it must be possible to express every matrix $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ as

$$c_1 \begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix} + c_2 \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix} + c_4 \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Equating corresponding entries on both sides yields linear systems

$$\begin{array}{rclcl} 3c_1 & + & 0c_2 & + & 0c_3 & + & 1c_4 & = & 0 \\ 6c_1 & - & 1c_2 & - & 8c_3 & + & 0c_4 & = & 0 \\ 3c_1 & - & 1c_2 & - & 12c_3 & - & 1c_4 & = & 0 \\ -6c_1 & + & 0c_2 & - & 4c_3 & + & 2c_4 & = & 0 \end{array} \quad \text{and} \quad \begin{array}{rclcl} 3c_1 & + & 0c_2 & + & 0c_3 & + & 1c_4 & = & a_{11} \\ 6c_1 & - & 1c_2 & - & 8c_3 & + & 0c_4 & = & a_{12} \\ 3c_1 & - & 1c_2 & - & 12c_3 & - & 1c_4 & = & a_{21} \\ -6c_1 & + & 0c_2 & - & 4c_3 & + & 2c_4 & = & a_{22} \end{array}$$

Since the coefficient matrix of both systems has determinant $\begin{vmatrix} 3 & 0 & 0 & 1 \\ 6 & -1 & -8 & 0 \\ 3 & -1 & -12 & -1 \\ -6 & 0 & -4 & 2 \end{vmatrix} = 48 \neq 0$, it

follows from parts (b), (e), and (g) of Theorem 2.3.8 that the homogeneous system has only the trivial solution and the nonhomogeneous system is consistent for all real values a_{11} , a_{12} , a_{21} , and a_{22} .

Therefore the matrices $\begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix}$, $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix}$, and $\begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$ are linearly independent and span M_{22} so that they form a basis for M_{22} .

6. Matrices $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, and $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ are linearly independent if the equation

$$c_1 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

has only the trivial solution. For these matrices to span M_{22} , it must be possible to express every matrix $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ as

$$c_1 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Equating corresponding entries on both sides in each equation yields linear systems

$$\begin{array}{rclcl} 1c_1 & + & 1c_2 & + & 0c_3 & + & 1c_4 & = & 0 \\ 1c_1 & - & 1c_2 & - & 1c_3 & + & 0c_4 & = & 0 \\ 1c_1 & + & 0c_2 & + & 1c_3 & + & 0c_4 & = & 0 \\ 1c_1 & + & 0c_2 & + & 0c_3 & + & 0c_4 & = & 0 \end{array} \quad \text{and} \quad \begin{array}{rclcl} 1c_1 & + & 1c_2 & + & 0c_3 & + & 1c_4 & = & a_{11} \\ 1c_1 & - & 1c_2 & - & 1c_3 & + & 0c_4 & = & a_{12} \\ 1c_1 & + & 0c_2 & + & 1c_3 & + & 0c_4 & = & a_{21} \\ 1c_1 & + & 0c_2 & + & 0c_3 & + & 0c_4 & = & a_{22} \end{array}$$

Since the coefficient matrix of both systems has determinant $\begin{vmatrix} 1 & 1 & 0 & 1 \\ 1 & -1 & -1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix} = 1 \neq 0$, it follows

from parts (b), (e), and (g) of Theorem 2.3.8 that the homogeneous system has only the trivial solution and the nonhomogeneous system is consistent for all real values a_{11} , a_{12} , a_{21} , and a_{22} .

Therefore the matrices $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, and $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ are linearly independent and span M_{22} so that they form a basis for M_{22} .

7. **(a)** Vectors $(2, -3, 1)$, $(4, 1, 1)$, and $(0, -7, 1)$ are linearly independent if the vector equation

$$c_1(2, -3, 1) + c_2(4, 1, 1) + c_3(0, -7, 1) = (0, 0, 0)$$

has only the trivial solution. This equation can be rewritten as a linear system

$$\begin{array}{rrcr} 2c_1 & + & 4c_2 & + & 0c_3 & = & 0 \\ -3c_1 & + & 1c_2 & - & 7c_3 & = & 0 \\ 1c_1 & + & 1c_2 & + & 1c_3 & = & 0 \end{array}$$

Since the determinant of the coefficient matrix of this system is $\begin{vmatrix} 2 & 4 & 0 \\ -3 & 1 & -7 \\ 1 & 1 & 1 \end{vmatrix} = 0$, it follows

from parts (b) and (g) of Theorem 2.3.8 that the homogeneous system has nontrivial solutions. Since the vectors $(2, -3, 1)$, $(4, 1, 1)$, and $(0, -7, 1)$ are linearly dependent, they do not form a basis for R^3 .

(b) Vectors $(1, 6, 4)$, $(2, 4, -1)$, and $(-1, 2, 5)$ are linearly independent if the vector equation

$$c_1(1, 6, 4) + c_2(2, 4, -1) + c_3(-1, 2, 5) = (0, 0, 0)$$

has only the trivial solution. This equation can be rewritten as a linear system

$$\begin{array}{rrcrcl} 1c_1 & + & 2c_2 & - & 1c_3 & = & 0 \\ 6c_1 & + & 4c_2 & + & 2c_3 & = & 0 \\ 4c_1 & - & 1c_2 & + & 5c_3 & = & 0 \end{array}$$

Since the determinant of the coefficient matrix of this system is $\begin{vmatrix} 1 & 2 & -1 \\ 6 & 4 & 2 \\ 4 & -1 & 5 \end{vmatrix} = 0$, it follows

from parts (b) and (g) of Theorem 2.3.8 that the homogeneous system has nontrivial solutions. Since the vectors $(1, 6, 4)$, $(2, 4, -1)$, and $(-1, 2, 5)$ are linearly dependent, they do not form a basis for R^3 .

8. Vectors $\mathbf{p}_1 = 1 - 3x + 2x^2$, $\mathbf{p}_2 = 1 + x + 4x^2$, and $\mathbf{p}_3 = 1 - 7x$ are linearly independent if the vector equation $c_1\mathbf{p}_1 + c_2\mathbf{p}_2 + c_3\mathbf{p}_3 = \mathbf{0}$ has only the trivial solution.

By grouping the terms on the left hand side as $c_1(1 - 3x + 2x^2) + c_2(1 + x + 4x^2) + c_3(1 - 7x) = (c_1 + c_2 + c_3) + (-3c_1 + c_2 - 7c_3)x + (2c_1 + 4c_2)x^2$ this equation can be rewritten as the linear system

$$\begin{array}{rrrrrr} c_1 & + & c_2 & + & c_3 & = & 0 \\ -3c_1 & + & c_2 & - & 7c_3 & = & 0 \\ 2c_1 & + & 4c_2 & & & = & 0 \end{array}$$

The coefficient matrix of this system has determinant $\begin{vmatrix} 1 & 1 & 1 \\ -3 & 1 & -7 \\ 2 & 4 & 0 \end{vmatrix} = 0$, thus it follows from

parts (b) and (g) of Theorem 2.3.8 that the homogeneous system has nontrivial solutions. Since the vectors $\mathbf{p}_1$, $\mathbf{p}_2$, and $\mathbf{p}_3$ are linearly dependent, we conclude that they do not form a basis for P_2 .

9. Matrices $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 2 & -2 \\ 3 & 2 \end{bmatrix}$, $\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$, and $\begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$ are linearly independent if the equation

$$c_1 \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 & -2 \\ 3 & 2 \end{bmatrix} + c_3 \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

has only the trivial solution. Equating corresponding entries on both sides yields a linear system

$$1c_1 + 2c_2 + 1c_3 + 0c_4 = 0$$

$$0c_1 - 2c_2 - 1c_3 - 1c_4 = 0$$

$$1c_1 + 3c_2 + 1c_3 + 1c_4 = 0$$

$$1c_1 + 2c_2 + 0c_3 + 1c_4 = 0$$

Since the determinant of the coefficient matrix of this system is $\begin{vmatrix} 1 & 2 & 1 & 0 \\ 0 & -2 & -1 & -1 \\ 1 & 3 & 1 & 1 \\ 1 & 2 & 0 & 1 \end{vmatrix} = 0$, it follows

from parts (b) and (g) of Theorem 2.3.8 that the homogeneous system has nontrivial solutions. Since the matrices $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 2 & -2 \\ 3 & 2 \end{bmatrix}$, $\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$, and $\begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$ are linearly dependent, we conclude that they do not form a basis for M_{22} .

11. (a) Expressing $\mathbf{w}$ as a linear combination of $\mathbf{u}_1$ and $\mathbf{u}_2$ we obtain

$$(1,1) = c_1(2,-4) + c_2(3,8)$$

Equating corresponding components on both sides yields the linear system

$$\begin{array}{rclcl} 2c_1 & + & 3c_2 & = & 1 \\ -4c_1 & + & 8c_2 & = & 1 \end{array}$$

whose augmented matrix has the reduced row echelon form $\begin{bmatrix} 1 & 0 & \frac{5}{28} \\ 0 & 1 & \frac{3}{14} \end{bmatrix}$. The solution of the linear system is $c_1 = \frac{5}{28}$, $c_2 = \frac{3}{14}$, therefore the coordinate vector is $(\mathbf{w})_S = (\frac{5}{28}, \frac{3}{14})$.

(b) Expressing $\mathbf{w}$ as a linear combination of $\mathbf{u}_1$ and $\mathbf{u}_2$ we obtain

$$(a,b) = c_1(1,1) + c_2(0,2)$$

Equating corresponding components on both sides yields the linear system

$$\begin{array}{rclcl} 1c_1 & + & 0c_2 & = & a \\ 1c_1 & + & 2c_2 & = & b \end{array}$$

whose augmented matrix has the reduced row echelon form $\begin{bmatrix} 1 & 0 & \frac{a}{2} \\ 0 & 1 & \frac{b-a}{2} \end{bmatrix}$. The solution of the linear system is $c_1 = a$, $c_2 = \frac{b-a}{2}$, therefore the coordinate vector is $(\mathbf{w})_S = (a, \frac{b-a}{2})$.

12. (a) Expressing $\mathbf{w}$ as a linear combination of $\mathbf{u}_1$ and $\mathbf{u}_2$ we obtain

$$(1,0) = c_1(1, -1) + c_2(1,1)$$

Equating corresponding components on both sides yields the linear system

$$\begin{array}{rrcr} c_1 & + & c_2 & = & 1 \\ -c_1 & + & c_2 & = & 0 \end{array}$$

whose augmented matrix has the reduced row echelon form $\begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \end{bmatrix}$. The solution of the linear system is $c_1 = \frac{1}{2}$, $c_2 = \frac{1}{2}$, therefore the coordinate vector is $(\mathbf{w})_S = (\frac{1}{2}, \frac{1}{2})$.

(b) Expressing $\mathbf{w}$ as a linear combination of $\mathbf{u}_1$ and $\mathbf{u}_2$ we obtain

$$(0,1) = c_1(1, -1) + c_2(1,1)$$

Equating corresponding components on both sides yields the linear system

$$\begin{array}{rrcr} c_1 & + & c_2 & = & 0 \\ -c_1 & + & c_2 & = & 1 \end{array}$$

whose augmented matrix has the reduced row echelon form $\begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} \end{bmatrix}$. The solution of the linear system is $c_1 = -\frac{1}{2}$, $c_2 = \frac{1}{2}$, therefore the coordinate vector is $(\mathbf{w})_S = (-\frac{1}{2}, \frac{1}{2})$.

13. (a) Expressing $\mathbf{v}$ as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ we obtain

$$(2, -1, 3) = c_1(1, 0, 0) + c_2(2, 2, 0) + c_3(3, 3, 3)$$

Equating corresponding components on both sides yields the linear system

$$\begin{array}{rcccccl} c_1 & + & 2c_2 & + & 3c_3 & = & 2 \\ & & 2c_2 & + & 3c_3 & = & -1 \\ & & & & 3c_3 & = & 3 \end{array}$$

which can be solved by back-substitution to obtain $c_3 = 1$, $c_2 = -2$, and $c_1 = 3$. The coordinate vector is $(\mathbf{v})_S = (3, -2, 1)$.

(b) Expressing $\mathbf{v}$ as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ we obtain

$$(5, -12, 3) = c_1(1, 2, 3) + c_2(-4, 5, 6) + c_3(7, -8, 9)$$

Equating corresponding components on both sides yields the linear system

$$\begin{array}{rcccccl} 1c_1 & - & 4c_2 & + & 7c_3 & = & 5 \\ 2c_1 & + & 5c_2 & - & 8c_3 & = & -12 \\ 3c_1 & + & 6c_2 & + & 9c_3 & = & 3 \end{array}$$

whose augmented matrix has the reduced row echelon form $\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$. The solution of the linear system is $c_1 = -2$, $c_2 = 0$, and $c_3 = 1$. The coordinate vector is $(\mathbf{v})_S = (-2, 0, 1)$.

14. (a) Since $\mathbf{p} = 4\mathbf{p}_1 + (-3)\mathbf{p}_2 + 1\mathbf{p}_3$ we conclude that the coordinate vector is $(\mathbf{p})_S = (4, -3, 1)$.

(b) Expressing $\mathbf{p}$ as a linear combination of $\mathbf{p}_1$, $\mathbf{p}_2$, and $\mathbf{p}_3$ we obtain

$$2 - x + x^2 = c_1(1 + x) + c_2(1 + x^2) + c_3(x + x^2)$$

Grouping the terms on the right hand side according to powers of x yields

$$2 - x + x^2 = (c_1 + c_2) + (c_1 + c_3)x + (c_2 + c_3)x^2$$

For this equality to hold for all real x , the coefficients associated with the same power of x on both sides must match. This leads to the linear system

$$\begin{array}{rclcl} c_1 & + & c_2 & & = & 2 \\ c_1 & & & + & c_3 & = & -1 \\ & & c_2 & + & c_3 & = & 1 \end{array}$$

whose augmented matrix has the reduced row echelon form $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix}$. The solution is $c_1 = 0$, $c_2 = 2$, $c_3 = -1$, therefore the coordinate vector is $(\mathbf{p})_S = (0, 2, -1)$.

15. Matrices (vectors in M_{22}) A_1, A_2, A_3 , and A_4 are linearly independent if the equation

$$k_1 A_1 + k_2 A_2 + k_3 A_3 + k_4 A_4 = \mathbf{0}$$

has only the trivial solution. For these matrices to span M_{22} , it must be possible to express every matrix $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ as

$$k_1 A_1 + k_2 A_2 + k_3 A_3 + k_4 A_4 = B$$

The left hand side of each of these equations is the matrix $\begin{bmatrix} k_1 & k_1 + k_2 \\ k_1 + k_2 + k_3 & k_1 + k_2 + k_3 + k_4 \end{bmatrix}$.

Equating corresponding entries, these two equations can be rewritten as linear systems

$$\begin{array}{rclcl} k_1 & & & = & 0 \\ k_1 & + & k_2 & & = & 0 \\ k_1 & + & k_2 & + & k_3 & = & 0 \\ k_1 & + & k_2 & + & k_3 & + & k_4 & = & 0 \end{array} \quad \text{and} \quad \begin{array}{rclcl} k_1 & & & = & a \\ k_1 & + & k_2 & & = & b \\ k_1 & + & k_2 & + & k_3 & = & c \\ k_1 & + & k_2 & + & k_3 & + & k_4 & = & d \end{array}$$

Since the coefficient matrix of both systems has determinant $\begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{vmatrix} = 1 \neq 0$, it follows from

parts (b), (e), and (g) of Theorem 2.3.8 that the homogeneous system has only the trivial solution and the nonhomogeneous system is consistent for all real values a, b, c and d . Therefore the matrices A_1, A_2, A_3 , and A_4 are linearly independent and span M_{22} so that they form a basis for M_{22} .

To express $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ as a linear combination of the matrices A_1, A_2, A_3 , and A_4 , we form the nonhomogeneous system as above, with the appropriate right hand side values

$$\begin{array}{rcl} k_1 & & = 1 \\ k_1 + k_2 & & = 0 \\ k_1 + k_2 + k_3 & & = 1 \\ k_1 + k_2 + k_3 + k_4 & = & 0 \end{array}$$

which can be solved by forward-substitution to obtain $k_1 = 1, k_2 = -1, k_3 = 1, k_4 = -1$.

This allows us to express $A = 1A_1 - 1A_2 + 1A_3 - 1A_4$.

The coordinate vector is $(A)_S = (1, -1, 1, -1)$.

16. Matrices (vectors in M_{22}) A_1, A_2, A_3 , and A_4 are linearly independent if the equation

$$k_1 A_1 + k_2 A_2 + k_3 A_3 + k_4 A_4 = \mathbf{0}$$

has only the trivial solution. For these matrices to span M_{22} , it must be possible to express every matrix $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ as

$$k_1 A_1 + k_2 A_2 + k_3 A_3 + k_4 A_4 = B$$

The left hand side of each of these equations is the matrix $\begin{bmatrix} k_1 + k_2 + k_3 & k_2 \\ k_1 + k_4 & k_3 \end{bmatrix}$. Equating corresponding entries, these two equations can be rewritten as linear systems

$$\begin{array}{rclcl} k_1 & + & k_2 & + & k_3 & & = & 0 \\ & & k_2 & & & & = & 0 \\ k_1 & & & & & + & k_4 & = & 0 \\ & & & & k_3 & & & = & 0 \end{array} \quad \text{and} \quad \begin{array}{rclcl} k_1 & + & k_2 & + & k_3 & & = & a \\ & & k_2 & & & & = & b \\ k_1 & & & & & + & k_4 & = & c \\ & & & & k_3 & & & = & d \end{array}$$

Since the coefficient matrix of both systems has determinant $\begin{vmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{vmatrix} = -1 \neq 0$, it follows from

parts (b), (e), and (g) of Theorem 2.3.8 that the homogeneous system has only the trivial solution and the nonhomogeneous system is consistent for all real values a, b, c and d . Therefore the matrices A_1, A_2, A_3 , and A_4 are linearly independent and span M_{22} so that they form a basis for M_{22} .

To express $A = \begin{bmatrix} 6 & 2 \\ 5 & 3 \end{bmatrix}$ as a linear combination of the matrices A_1, A_2, A_3 , and A_4 , we form the nonhomogeneous system as above, with the appropriate right hand side values

$$\begin{array}{rclcl} k_1 & + & k_2 & + & k_3 & & = & 6 \\ & & k_2 & & & & = & 2 \\ k_1 & & & & & + & k_4 & = & 5 \\ & & & & k_3 & & & = & 3 \end{array}$$

The augmented matrix of this system has the reduced row echelon form $\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{bmatrix}$ therefore

the solution is $k_1 = 1, k_2 = 2, k_3 = 3, k_4 = 4$.

This allows us to express $A = 1A_1 + 2A_2 + 3A_3 + 4A_4$. The coordinate vector is $(A)_S = (1, 2, 3, 4)$.

17. Vectors $\mathbf{p}_1, \mathbf{p}_2$, and $\mathbf{p}_3$ are linearly independent if the vector equation

$$c_1\mathbf{p}_1 + c_2\mathbf{p}_2 + c_3\mathbf{p}_3 = \mathbf{0}$$

has only the trivial solution. For these vectors to span P_2 , it must be possible to express every vector $\mathbf{p} = a_0 + a_1x + a_2x^2$ in P_2 as

$$c_1\mathbf{p}_1 + c_2\mathbf{p}_2 + c_3\mathbf{p}_3 = \mathbf{p}$$

Grouping the terms on the left hand sides as $c_1(1 + x + x^2) + c_2(x + x^2) + c_3x^2 = c_1 + (c_1 + c_2)x + (c_1 + c_2 + c_3)x^2$ these two equations can be rewritten as linear systems

$$\begin{array}{rclclcl} c_1 & & & = & 0 & & c_1 & & & = & a_0 \\ c_1 & + & c_2 & = & 0 & \text{and} & c_1 & + & c_2 & = & a_1 \\ c_1 & + & c_2 & + & c_3 & = & 0 & & c_1 & + & c_2 & + & c_3 & = & a_2 \end{array}$$

Since the coefficient matrix of both systems has determinant $\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = 1 \neq 0$, it follows from

parts (b), (e), and (g) of Theorem 2.3.8 that the homogeneous system has only the trivial solution and the nonhomogeneous system is consistent for all real values a_0, a_1 , and a_2 . Therefore the vectors $\mathbf{p}_1, \mathbf{p}_2$, and $\mathbf{p}_3$ are linearly independent and span P_2 so that they form a basis for P_2 .

To express $\mathbf{p} = 7 - x + 2x^2$ as a linear combination of the vectors $\mathbf{p}_1, \mathbf{p}_2$, and $\mathbf{p}_3$, we form the nonhomogeneous system as above, with the appropriate right hand side values

$$\begin{array}{rclclcl} c_1 & & & = & 7 \\ c_1 & + & c_2 & = & -1 \\ c_1 & + & c_2 & + & c_3 & = & 2 \end{array}$$

which can be solved by forward-substitution to obtain $c_1 = 7, c_2 = -8, c_3 = 3$.

This allows us to express $\mathbf{p} = 7\mathbf{p}_1 - 8\mathbf{p}_2 + 3\mathbf{p}_3$. The coordinate vector is $(\mathbf{p})_S = (7, -8, 3)$.

18. Vectors $\mathbf{p}_1$, $\mathbf{p}_2$, and $\mathbf{p}_3$ are linearly independent if the vector equation

$$c_1\mathbf{p}_1 + c_2\mathbf{p}_2 + c_3\mathbf{p}_3 = \mathbf{0}$$

has only the trivial solution. For these vectors to span P_2 , it must be possible to express every vector $\mathbf{p} = a_0 + a_1x + a_2x^2$ in P_2 as

$$c_1\mathbf{p}_1 + c_2\mathbf{p}_2 + c_3\mathbf{p}_3 = \mathbf{p}$$

Grouping the terms on the left hand sides as $c_1(1 + 2x + x^2) + c_2(2 + 9x) + c_3(3 + 3x + 4x^2) = (c_1 + 2c_2 + 3c_3) + (2c_1 + 9c_2 + 3c_3)x + (c_1 + 4c_3)x^2$ these two equations can be rewritten as linear systems

$$\begin{array}{rclcl}
 c_1 & + & 2c_2 & + & 3c_3 & = & 0 \\
 2c_1 & + & 9c_2 & + & 3c_3 & = & 0 \\
 c_1 & & & + & 4c_3 & = & 0
 \end{array}
 \quad \text{and} \quad
 \begin{array}{rclcl}
 c_1 & + & 2c_2 & + & 3c_3 & = & a_0 \\
 2c_1 & + & 9c_2 & + & 3c_3 & = & a_1 \\
 c_1 & & & + & 4c_3 & = & a_2
 \end{array}$$

Since the coefficient matrix of both systems has determinant $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 9 & 3 \\ 1 & 0 & 4 \end{vmatrix} = -1 \neq 0$, it follows from

parts (b), (e), and (g) of Theorem 2.3.8 that the homogeneous system has only the trivial solution and the nonhomogeneous system is consistent for all real values a_0 , a_1 , and a_2 . Therefore the vectors $\mathbf{p}_1$, $\mathbf{p}_2$, and $\mathbf{p}_3$ are linearly independent and span P_2 so that they form a basis for P_2 .

To express $\mathbf{p} = 2 + 17x - 3x^2$ as a linear combination of the vectors $\mathbf{p}_1$, $\mathbf{p}_2$, and $\mathbf{p}_3$, we form the nonhomogeneous system as above, with the appropriate right hand side values

$$\begin{array}{rclcl}
 c_1 & + & 2c_2 & + & 3c_3 & = & 2 \\
 2c_1 & + & 9c_2 & + & 3c_3 & = & 17 \\
 c_1 & & & + & 4c_3 & = & -3
 \end{array}$$

The augmented matrix of this system has the reduced row echelon form $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix}$ therefore

the solution is $c_1 = 1$, $c_2 = 2$, $c_3 = -1$. This allows us to express $\mathbf{p} = 1\mathbf{p}_1 + 2\mathbf{p}_2 + (-1)\mathbf{p}_3$.

The coordinate vector is $(\mathbf{p})_S = (1, 2, -1)$.

Thank You